Project Cardiovascular Disease Data Visualization Using Python, Power BI and Tableau

AIM

To analyze the Cardio dataset and create three visualizations - bar chart, line chart, and pie chart - using Python, Power BI, and Tableau to understand the occurrence of cardiovascular disease, identify risk factors, and analyze patient distribution.

PROBLEM STATEMENT

Hospitals collect large amounts of patient data such as age, blood pressure, cholesterol, and lifestyle details. Understanding this data manually is difficult. This project uses Python libraries such as Pandas and Matplotlib, along with Power BI and Tableau, to process the Cardio dataset and create visualizations. These visualizations help identify age groups with more cardio cases, observe blood pressure trends with age, and understand the proportion of patients with cardiovascular disease.

SOFTWARE REQUIREMENTS

- Platform: Google Colab, Power BI Desktop, Tableau Desktop
- Programming Language: Python
- Dataset Format: CSV (.csv)
- Libraries: Pandas, Matplotlib
- Input Dataset: Cardio CSV file (Cardio_Dataset.csv)

DATASET DESCRIPTION

The Cardio dataset contains 700 patient records with the following columns.

Column Name	Description
id	Unique ID of the patient
age	Age of the patient in years
gender	Gender of the patient (Male / Female)
height, weight	Height (cm) and weight (kg) of the patient
ap_hi, ap_lo	Systolic and diastolic blood pressure
cholesterol	Cholesterol level (Normal / Above Normal / High)
glucose	Glucose level (Normal / Above Normal / High)
smoke, alco, active	Smoking, alcohol intake and physical activity (0 = No, 1 = Yes)
cardio	Presence of cardiovascular disease (0 = No, 1 = Yes)

1. PYTHON

PROCEDURE

1. Import the dataset into Python, Power BI, and Tableau.

2. Inspect the dataset and identify the age, blood pressure, cholesterol, and cardio columns.
3. Handle missing values and ensure that dates and numerical columns have the correct data types.
4. Create an Age Group column from the age column and calculate BMI from height and weight.
5. Create a bar chart to compare values across categories.
6. Create a line chart to analyze trends.
7. Create a pie chart to show the distribution across categories.
8. Use Pandas and Matplotlib to generate the three visualizations in Python.
9. Import the dataset into Power BI and Tableau and recreate the same visualizations.
10. Compare the charts and identify trends, popular categories, and business insights.

Program and Output

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the dataset
df = pd.read_csv("Cardio_Dataset.csv")

# Clean column names
df.columns = df.columns.str.strip()

# Convert required columns
for col in ["age", "ap_hi", "ap_lo", "cardio"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

# Remove missing values
df = df.dropna(subset=["age", "ap_hi", "cholesterol", "cardio"])

# Create age groups
df["Age Group"] = pd.cut(df["age"], bins=[29, 39, 49, 59, 100],
                         labels=["30-39", "40-49", "50-59", "60+"])

# 1. BAR CHART - Cardio Cases by Age Group
age_cases = df[df["cardio"] == 1].groupby("Age Group", observed=True).size()

plt.figure(figsize=(8, 5))
age_cases.plot(kind="bar", color="crimson")
plt.title("Cardio Cases by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Patients")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

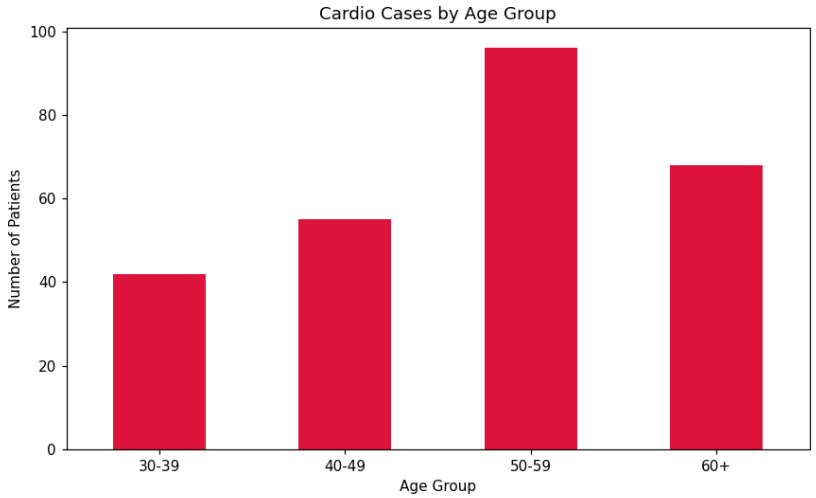


Fig 1: Bar chart - Cardio Cases by Age Group

```
# 2. LINE CHART - Average Systolic BP by Age
bp_trend = df.groupby("age")["ap_hi"].mean()

plt.figure(figsize=(9, 5))
plt.plot(bp_trend.index, bp_trend.values, marker="o", color="darkblue")
plt.title("Average Systolic Blood Pressure by Age")
plt.xlabel("Age")
plt.ylabel("Average Systolic BP (ap_hi)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

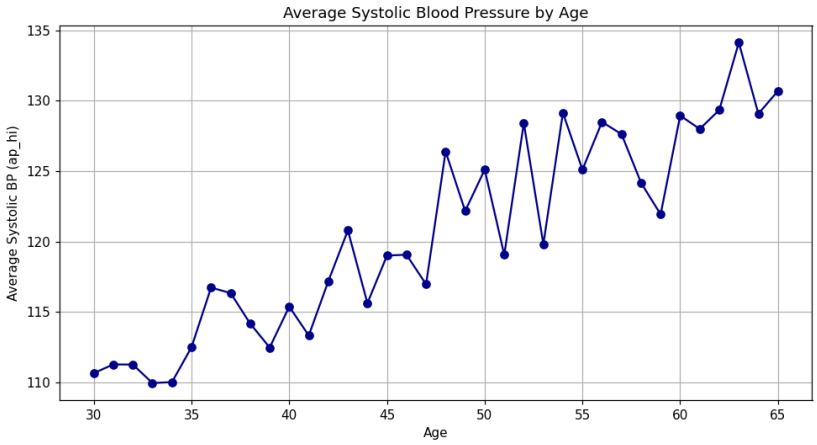


Fig 2: Line chart - Average Systolic BP by Age

```
# 3. PIE CHART - Cardio Disease Distribution
cardio_dist = df["cardio"].value_counts().sort_index()

plt.figure(figsize=(6, 6))
plt.pie(
    cardio_dist.values,
    labels=["No Disease", "Disease"],
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Cardio Disease Distribution")
plt.tight_layout()
plt.show()
```

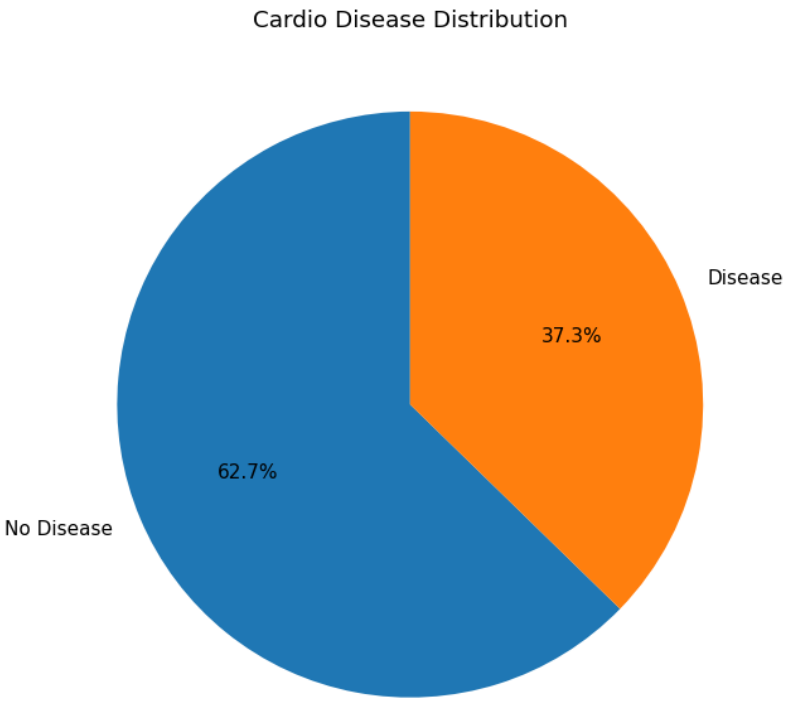


Fig 3: Pie chart - Cardio Disease Distribution

2. POWER BI

1. Open Power BI Desktop and select Home > Get Data > Text/CSV.
2. Select the Cardio_Dataset.csv file and click Load.
3. Check the data types of age, ap_hi, cholesterol, and cardio.
4. Create a calculated column using Table tools > New column.
   Age Group = IF('Cardio'[age] >= 50, "50+", "Below 50")
5. Create a bar chart by placing Age Group on the Y-axis and Count of id on the X-axis, with a visual-level filter cardio = 1.
6. Create a line chart using age on the X-axis and Average of ap_hi on the Y-axis.
7. Create a pie chart using cardio in Legend and Count of id in Values.
8. Add Card visuals to display Total Patients, Average Age, and Cardio Cases.
9. Add a gender or cholesterol slicer to filter the report.
10. Arrange the visuals, format titles and colors, and save the report as Cardio.pbix.



Fig: Cardio Analytics Dashboard (Power BI)

3. TABLEAU

1. Open Tableau Desktop.
2. Under Connect, select Text File and open Cardio_Dataset.csv.
3. Open a new worksheet.
4. Create the bar chart by placing Age Group (calculated field) on Columns and Count of id on Rows, and drag cardio to Color.
5. For the line chart, use age on Columns and Average of ap_hi on Rows, then select Line in the Marks card.
6. Create a pie chart by selecting Pie in the Marks card, placing cholesterol on Color and Count of id on Angle.
7. Set the aggregation to Average or Count where appropriate.
8. Add chart titles and customize the colors.
9. Select New Dashboard and drag all the worksheets onto the dashboard.
10. Save the workbook as Cardio.twbx or .twb.

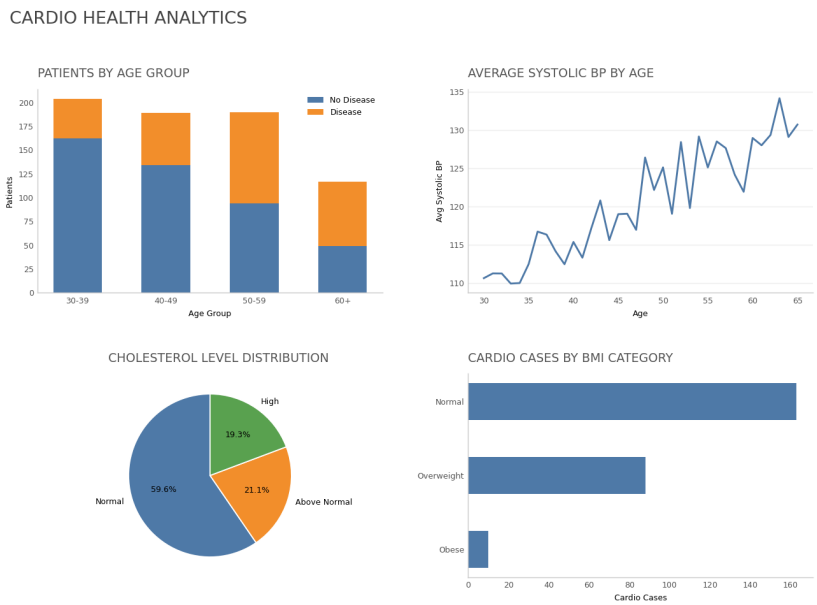


Fig: Cardio Health Analytics Dashboard (Tableau)

VISUALIZATIONS

1. Bar Chart - Cardio Cases by Age Group

Compares the number of patients with cardiovascular disease across age groups.

2. Line Chart - Average Systolic BP by Age

Shows how average systolic blood pressure changes as age increases.

3. Pie Chart - Cardio Disease Distribution

Displays the proportion of patients with and without cardiovascular disease.

OBSERVATION AND INSIGHTS

- The age group 50-59 has the highest number of cardio cases (96), and the disease rate rises from 20.6% in the 30-39 group to 58.1% in the 60+ group.
- Average systolic blood pressure increases with age, from about 111 mmHg at ages 30-34 to about 130 mmHg at ages 61-65.
- 37.3% of the 700 patients have cardiovascular disease.
- Patients with High cholesterol show a disease rate of 43.0%, compared with 34.1% for Normal cholesterol.
- Python, Power BI, and Tableau provide different ways to create and present the same visual analysis.

RESULT

The Cardio dataset was used to create bar charts, line charts, and pie charts using Python, Power BI, and Tableau. The visualizations support analysis of cardiovascular disease across age groups, blood pressure trends, and patient distribution using three different data visualization tools.